Viren Khandal

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Experience

- **Tesla** — Fremont, CA
  Senior Machine Learning Engineer — *Dec 2022 – Present*
  - Leading all ML efforts across vision-guided robotics, large-scale automated manufacturing defect detection, and core ML for an internal computer vision training platform.
  - Designed and deployed production computer vision systems for automated quality inspection across Tesla's high-throughput manufacturing lines.
  - Built real-time inference pipelines optimized for low-latency, low-intervention production environments, reducing manual inspection overhead at scale.
- **Ponder (Stealth)** — Remote
  Co-Founder — *Oct 2024 – Jan 2026*
  - Led full backend and ML development for an AI app that generates multi-perspective audio podcasts from any user query, with each perspective grounded in cited research and credible sources.
  - Architected retrieval-augmented generation pipelines to synthesize web-scale information into structured, multi-viewpoint audio content with source attribution.
  - Built end-to-end backend infrastructure including API design, audio generation, and serving systems to support real-time podcast creation at scale.
- **University of California, Berkeley – VeHICal** — Berkeley, CA
  Research Engineer — *May 2021 – Dec 2022*
  - Led research on optimizing human performance during autonomous-to-human perception handoffs in autonomous vehicles under Professors Sanjit Seshia and Bjoern Hartmann (vehical.org).
  - Developed Virtual Reality (VR) tools to conduct human-in-the-loop experiments with high ecological validity for studying driver takeover scenarios.
  - Formalized stochastic models for safe perception handoffs, contributing to verified human-autonomy interaction frameworks.
- **Stanford Autonomous Systems Lab** — Stanford, CA
  Visiting Researcher — *Apr 2022 – Dec 2022*
  - Investigated effects of out-of-distribution (OOD) events on human-autonomy interaction in autonomous driving under Professor Marco Pavone (stanfordasl.github.io).
  - Developed a framework enabling safe real-time control handoffs from autonomous vehicle to human driver during OOD events.

Education

- **University of California, Berkeley** — Berkeley, CA
  B.A. Computer Science & Applied Mathematics (Double Major) — *Aug 2019 – May 2022*
  - Specialization in Machine Learning and Artificial Intelligence
  - Coursework: Deep Learning, Machine Learning, Artificial Intelligence, Efficient Algorithms, Data Structures, Statistical Prediction, Numerical/Real/Complex Analysis, Advanced Linear Algebra, Abstract Algebra
- **Stanford University** — Palo Alto, CA
  Certificate in Computer Science — *May 2018 – Sept 2018*
  - Coursework: Data Mining and Analysis, Client-Side Internet Technologies

Skills

- **Languages**: Python, C/C++, Go, Java, JavaScript, SQL, Shell
- **ML/AI**: PyTorch, TensorFlow, OpenCV, ONNX, TensorRT, Hugging Face, scikit-learn
- **Infrastructure**: AWS, GCP, Docker, Kubernetes, Git, CI/CD, Linux
- **Domains**: Computer Vision, Object Detection, Manufacturing ML, Autonomous Driving, Human-Autonomy Interaction